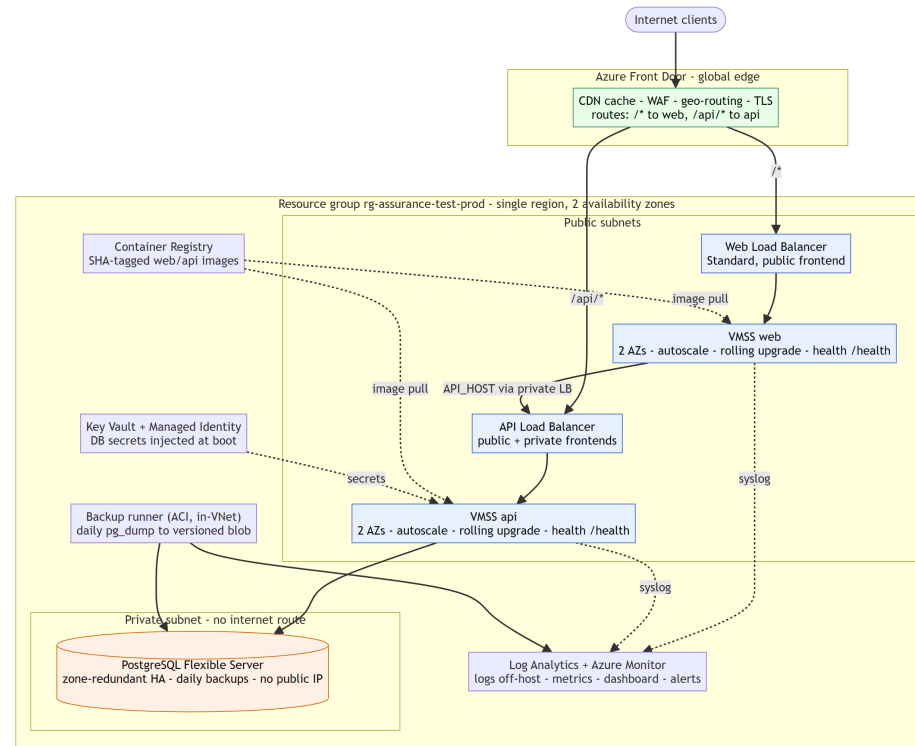


Continuous delivery for a 3-tier Node platform

Scalable, secure, self-healing, zero-downtime — on Azure, fully IaC

Architecture

assurance-test - 3-tier continuous delivery on Azure



Azure Front Door -> VMSS web/api (2 AZs) -> private PostgreSQL (zone-redundant HA)

How each requirement is met

Requirement	Implementation
Web + API on the Internet	Front Door routes /* and /api/* to per-tier VMSS behind load balancers
DB not public	PostgreSQL Flexible Server, private VNet access, NSG-restricted
Fully IaC	Terraform, remote state with locking, one resource group
Survive instance failure	VMSS across 2 AZs, health probes, auto-repair; zone-redundant DB HA
Update without downtime	Health-gated VMSS rolling upgrades, SHA-tagged images
Automated deployment	GitHub Actions: test -> build -> apply -> rolling -> smoke
Daily backups	Flexible Server PITR + in-VNet pg_dump to versioned blob
Logs off-host	Azure Monitor Agent + diagnostics -> Log Analytics
Historical metrics	Azure Monitor dashboard + alerts
CDN by location	Azure Front Door global edge + static caching

Delivery pipeline & resilience

- **Pipeline:** push -> lint/test -> build+push SHA-tagged images to ACR -> terraform apply -> health-gated rolling upgrade -> smoke test through Front Door
- **Zero downtime:** a new image only replaces an instance after the new one passes `/health` ; Front Door and the LB route only to healthy nodes
- **Self-healing:** unhealthy instances are auto-repaired; the DB fails over to a standby in another zone
- **Secrets:** Key Vault + managed identity, no credentials on disk or in git
- **State:** shared Terraform state in Azure Storage with blob-lease locking (team-ready)

One app, three deployment models

- **VMSS (reference):** VM Scale Sets running containers — the deployed, verified primary
- **AKS + Helm:** the same app on Kubernetes — Deployments, HPA, rolling updates, zone spread
- **Azure Container Apps:** serverless containers with built-in ingress, revisions, scale-to-zero

Deliberately chose VMSS as the primary: it maps directly to "start/stop/scale nodes" and keeps the moving parts minimal, while the alternatives show the same architecture on other platforms.